



MANIPAL

ACADEMY *of* HIGHER EDUCATION

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MANIPAL SCHOOL OF INFORMATION SCIENCES

(A Constituent unit of MAHE, Manipal)

CYE 5103 Linux OS and Scripting – Class 1

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Part A — User and System Investigation (10 minutes)

Determine the username of the account you are currently using.

whoami

Display your UID, primary GID, and supplementary groups.

```

haris@ubuntu-arm: ~/Documents/LoS/classwork
~/Documents/LoS/classwork

haris@ubuntu-arm:~/Documents/LoS/classwork$ id
uid=1000(haris) gid=1000(haris) groups=1000(haris),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),100(users),111(lpadmin),114(lxd)
haris@ubuntu-arm:~/Documents/LoS/classwork$ 

```

Display all users currently logged in to the system.

```

haris
haris@ubuntu-arm:~/Documents/LoS/classwork$ w
 09:27:00 up 10 min,  1 user,  load average: 0.03, 0.08, 0.08
USER  TTY      FROM             LOGIN@   IDLE   JCPU   PCPU   WHAT
haris  tty2      -                09:16   10:36   0.00s  ?      /usr/libexec/gnome-session-init-worker ubuntu
haris@ubuntu-arm:~/Documents/LoS/classwork$ who
haris@ubuntu-arm:~/Documents/LoS/classwork$ 

```

Determine what activities the currently logged-in users are performing.

Examine /etc/passwd and identify the following information for your current account:

Username

UID

GID

Home directory

Login shell

```

pipewire:x:981:981:system user for pipewire:/nonexistent:/usr/sbin/nologin
hplip:x:111:7:HPLIP system user:/run/hplip:/bin/false
gnome-remote-desktop:x:980:980:GNOME Remote Desktop:/var/lib/gnome-remote-desktop:/usr/sbin/nologin
polkitd:x:979:979:User for polkitd:/usr/sbin/nologin
rtkit:x:978:978:RealtimeKit:/proc:/usr/sbin/nologin
colord:x:977:977:colord colour management daemon:/var/lib/colord:/usr/sbin/nologin
gdm:x:975:975:Gnome Display Manager:/var/lib/gdm3:/bin/false
haris:x:1000:1000:Haris:/home/haris:/bin/bash
haris@ubuntu-arm:~/Documents/LoS/classwork$ 

```

Determine how many user-account entries are present in /etc/passwd.

```

grep -c '/bin/(bash|sh|zsh)$' /etc/passwd
2
haris@ubuntu-arm:~/Documents/LoS/classwork$ clear

```

Display only the username and UID fields from /etc/passwd.

```
haris@ubuntu-arm:~/Documents/LoS/classwork$ cut -d: -f1,3 /etc/passwd
root:0
daemon:1
bin:2
sys:3
sync:4
games:5
man:6
lp:7
mail:8
news:9
uucp:10
proxy:13
www-data:33
backup:34
list:38
irc:39
_apt:42
nobody:65534
systemd-network:998
dhcpcd:996
messagebus:995
syslog:100
systemd-resolve:989
_chrony:988
tss:987
uuid:101
systemd-oom:986
whoopsie:102
dnsmasq:999
avahi:103
nm-openvpn:984
tcpdump:983
sssd:104
speech-dispatcher:105
usbmux:106
cups-pk-helper:107
fwupd-refresh:982
saned:108
geoclue:109
cups-browsed:110
pipewire:981
hplip:111
gnome-remote-desktop:980
polkitd:979
rtkit:978
colord:977
gdm:975
haris:1000
haris@ubuntu-arm:~/Documents/LoS/classwork$
```

Find all accounts whose UID is 0. Explain the security significance of UID 0.

```
grep: command not found
haris@ubuntu-arm:~/Documents/LoS/classwork$ egrep '^[^:]*:[^:]*:0:' /etc/passwd
root:x:0:0:root:/root:/bin/bash
haris@ubuntu-arm:~/Documents/LoS/classwork$
```

- UID 0 is the superuser or root identifier. It carries total power and control over the operating system.
- A user with UID 0 bypasses all standard file permissions and security restrictions. They can read, write, or delete any file on the disk.

Compare /etc/passwd and /etc/shadow. Why are password hashes stored in /etc/shadow instead of /etc/passwd?

- */etc/shadow is a secure, root-only readable file that stores sensitive password hashes and password whereas, /etc/passwd must remain public so system utilities can map user IDs to names*

Part B — Creating and Verifying Users (20 Minutes)

Create the following users and ensure that a home directory is created for each:

alice, bob, charlie

```
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
Sorry, passwords do not match.
passwd: Have exhausted maximum number of retries for service
passwd: password unchanged
warn: '/bin/passwd alice' failed with status 10. Continuing.
warn: wrong password given or password retyped incorrectly
Try again? [y/N] y
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for alice
Enter the new value, or press ENTER for the default
    Full Name []: alice
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] y
haris@ubuntu-arm:~/Documents/LoS/classes$ sudo adduser bob
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for bob
Enter the new value, or press ENTER for the default
    Full Name []: bob
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] y
haris@ubuntu-arm:~/Documents/LoS/classes$ sudo adduser charlie
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for charlie
Enter the new value, or press ENTER for the default
    Full Name []: charlie
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] y
haris@ubuntu-arm:~/Documents/LoS/classes$
```

Verify that all three accounts have been added to /etc/passwd.

```
colord:x:977:977:colord colour management daemon:/var/lib/colord:/usr/sbin/nologin
gdm:x:975:975:Gnome Display Manager:/var/lib/gdm3:/bin/false
haris:x:1000:1000:Haris:/home/haris:/bin/bash
alice:x:1001:1001:alice,,,:/home/alice:/bin/bash
bob:x:1002:1002:bob,,,:/home/bob:/bin/bash
charlie:x:1003:1003:charlie,,,:/home/charlie:/bin/bash
haris@ubuntu-arm:~/Documents/LoS/classes$
```

Determine the UID assigned to each newly created user.

```
haris@ubuntu-arm:~/Documents/LoS/classes$ id bob
uid=1002(bob) gid=1002(bob) groups=1002(bob),100(users)
haris@ubuntu-arm:~/Documents/LoS/classes$ id charlie
uid=1003(charlie) gid=1003(charlie) groups=1003(charlie),100(users)
haris@ubuntu-arm:~/Documents/LoS/classes$ id alice
uid=1001(alice) gid=1001(alice) groups=1001(alice),100(users)
haris@ubuntu-arm:~/Documents/LoS/classes$
```

Determine the home directory and default shell assigned to alice.

```
alice:x:1001:1001:alice,,,:/home/alice:/bin/bash
bob:x:1002:1002:bob,,,:/home/bob:/bin/bash
charlie:x:1003:1003:charlie,,,:/home/charlie:/bin/bash
haris@ubuntu-arm:~/Documents/LoS/Classes$
```

Inspect /home and verify that the required home directories were created.

```
haris@ubuntu-arm:~/Documents/LoS/Classes$ ls /home/
alice bob charlie haris
haris@ubuntu-arm:~/Documents/LoS/Classes$
```

Set passwords for all three accounts.

Switch from your current account to alice using a login shell. Verify the change using an appropriate command, and then return to your original account.

```
alice bob charlie haris
haris@ubuntu-arm:~/Documents/LoS/Classes$ su - alice
Password:
alice@ubuntu-arm:~$ su
Password:
su: Authentication failure
alice@ubuntu-arm:~$ ^C
alice@ubuntu-arm:~$ su - haris
Password:
haris@ubuntu-arm:~$
```

Part C — Group Management and Access Control (30 Minutes)

Create two groups:

security and developers

Add alice and charlie to the security group.

Add bob to the developers group.

Without switching accounts, verify the group membership of all three users.

```
haris@ubuntu-arm:~$ sudo groupadd developers
haris@ubuntu-arm:~$ sudo groupadd security
haris@ubuntu-arm:~$ sudo usermod -aG security alice
haris@ubuntu-arm:~$ sudo usermod -aG security charlie
haris@ubuntu-arm:~$ sudo usermod -aG developers bob
haris@ubuntu-arm:~$ groups alice
alice : alice users security
haris@ubuntu-arm:~$ groups charlie
charlie : charlie users security
haris@ubuntu-arm:~$ groups bob
bob : bob users developers
haris@ubuntu-arm:~$
```

Display the entries corresponding to security and developers from /etc/group.

```
haris@ubuntu-arm:~$ egrep 'security|developers' /etc/group
developers:x:1004:bob
security:x:1005:alice,charlie
haris@ubuntu-arm:~$
```

Add bob to the security group without removing him from the developers group.
Verify that bob now belongs to both groups.

```
haris@ubuntu-arm:~$ sudo usermod -aG security bob
haris@ubuntu-arm:~$ groups bob
bob : bob users developers security
haris@ubuntu-arm:~$
```